



OOPS ASSIGNMENTS

SECTION A: POLYMORPHISM (4 Assignments)

Assignment 2.1

Create method overloading using a class Messenger with different send() methods.

Assignment 2.2

Create method overriding using Vehicle → Bike.

Assignment 2.3

Implement polymorphism — create multiple classes with method overloading.

Assignment 2.4

Implement polymorphism — create multiple classes with method overriding.

SECTION B: KEYWORDS (this & super) (2 Assignments)

Assignment 2.5

Use this keyword to differentiate instance and local variables.

Assignment 2.6

Use super keyword to call parent constructor.

SECTION C: ABSTRACT CLASSES (4 Assignments)

Assignment 2.7

Create an abstract class Remote and implement it in ACRemote.

Assignment 2.8

Create abstract class Exam and extend it in OnlineExam.

Assignment 2.9

Implement abstract classes — declare abstract & non-abstract methods.

Assignment 2.10

Define abstract methods inside a child class — use them.

SECTION D: INTERFACES (4 Assignments)

Assignment 2.11

Create interface Playable and implement it in MusicPlayer.

Assignment 2.12

Create an interface — connect it to a class — find difference between abstract class & interface.

Assignment 2.13

Create multiple inheritance using two interfaces.

Assignment 2.14

Understand interfaces & abstract classes with real examples.

SECTION E: ENCAPSULATION (3 Assignments)

Assignment 2.15

Implement encapsulation — make variables & methods private.

Assignment 2.16

Create setters & getters — access private variables from outside.

Assignment 2.17

Understand encapsulation effects with a real-time example.

SECTION F: EXCEPTION HANDLING (3 Assignments)

Assignment 2.18

Research exception handling in Java & use if-else to handle divide-by-zero.

Assignment 2.19

Use try-catch to handle divide-by-zero — compare with if-else.

Assignment 2.20

Create custom exceptions & throw them using throw keyword.
